

THOMAS JACOB MARANZATTO

<https://tmaranzatto.github.io/>

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APPOINTMENTS

Postdoctoral Fellow, University of Maryland Department of Electrical and Computer Engineering	2024-Present
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EDUCATION

PhD, University of Illinois Chicago (Advised by Lev Reyzin) Department of Mathematics, Statistics, and Computer Science	2020 - 2024
BA, New College of Florida Computer Science Major, Mathematics Minor	2016 - 2020

JOURNAL PUBLICATIONS

1. Arunabh Srivastava, **Thomas Maranzatto**, Sennur Ulukus. “Scalability and Freshness in Time-Varying Gossip Networks”. IEEE Transactions on Information Theory. Accepted in revision, 2026.
2. **Thomas Maranzatto** and Marcus Michelen “Age of Gossip From Connective Properties via First Passage Percolation,” in IEEE Transactions on Information Theory, vol. 71, no. 11, pp. 9019-9027, Nov. 2025, doi: 10.1109/TIT.2025.3612162.
3. Shreya Meel, Xiangliang Kong, **Thomas Maranzatto**, Itzhak Tamo, and Sennur Ulukus. “New Capacity Bounds for PIR on Graph and Multigraph-Based Replicated Storage,” in IEEE Transactions on Information Theory, vol. 72, no. 1, pp. 691-709, Jan. 2026, doi: 10.1109/TIT.2025.3631441.

CONFERENCE PUBLICATIONS

1. **Thomas Maranzatto** and Sennur Ulukus, “Deriving Moments in the Age of Gossip Process from Percolation,” IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Barcelona, Spain, 2026, pp. 21651-21655, doi: 10.1109/ICASSP55912.2026.11463463.
2. Arunabh Srivastava, **Thomas Maranzatto**, Sennur Ulukus. “Information Freshness in Dynamic Gossip Networks,” IEEE Information Theory Workshop (ITW), Sydney, Australia, 2025, pp. 1-6, doi: 10.1109/ITW62417.2025.11240391.
3. **Thomas Maranzatto**, Arunabh Srivastava and Sennur Ulukus, “Information Degradation and Misinformation in Gossip Networks,” IEEE International Symposium on Information Theory (ISIT), Ann Arbor, MI, USA, 2025, pp. 1-6, doi: 10.1109/ISIT63088.2025.11195622.
4. Shreya Meel, Xiangliang Kong, **Thomas Maranzatto**, Itzhak Tamo, and Sennur Ulukus. “Private Information Retrieval on Multigraph-Based Replicated Storage,” IEEE International Symposium on Information Theory (ISIT), Ann Arbor, MI, USA, 2025, pp. 1-6, doi: 10.1109/ISIT63088.2025.11195342.
5. Arunabh Srivastava, **Thomas Maranzatto**, Sennur Ulukus. “Age of Gossip with Time-Varying Topologies,” IEEE International Symposium on Information Theory (ISIT), Ann Arbor, MI, USA, 2025, pp. 1-6, doi: 10.1109/ISIT63088.2025.11195490.
6. Arunabh Srivastava, **Thomas Maranzatto**, Sennur Ulukus. “Age of Gossip with the Push-Pull Protocol,” IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP), Hyderabad, India, 2025, pp. 1-5, doi: 10.1109/ICASSP49660.2025.10890446.

7. **Thomas Maranzatto**. “Tree Trace Reconstruction - Reductions to String Trace Reconstruction,” IEEE International Symposium on Information Theory (ISIT), Athens, Greece, 2024, pp. 885-890, doi: 10.1109/ISIT57864.2024.10619306.
8. **Thomas Maranzatto**. “Age of Gossip in Random and Bipartite Networks,” IEEE International Symposium on Information Theory (ISIT), Athens, Greece, 2024, pp. 1173-1178, doi: 10.1109/ISIT57864.2024.10619401.
9. Xing Gao, **Thomas Maranzatto**, Lev Reyzin. “A Unified Analysis of Dynamic Interactive Learning,” 59th Annual Allerton Conference on Communication, Control, and Computing (Allerton), Monticello, IL, USA, 2023, pp. 1-8, doi: 10.1109/Allerton58177.2023.10313487.
10. Tania Roy, **Thomas Maranzatto**, Zachary Loomas. “Using machine learning algorithms (supervised) to generate automatically labeled dataset for detecting digital dating abuse from text messages,” FLAIRS, vol. 36, no. 1, May 2023, DOI: <https://doi.org/10.32473/flairs.36.133332>.
11. Tori Kraj, **Thomas Maranzatto**, Reynold Bailey, Cecilia Ovesdotter Alm, Joe Geigel. “Evaluating Audience Engagement of an Immersive Performance on a Virtual Stage,” Frameless: Vol. 2: Iss. 1, Article 4, 2020 Available at: <https://repository.rit.edu/frameless/vol2/iss1/4>

PRE-PRINTS

1. Sixu Li, **Thomas Maranzatto**, Jan Peszek, Trevor Teolis, Semih Akkoc, Konstantin Riedl, Sennur Ulukus, Nicolás García Trillos. “On the Diverse Dynamical Behaviors Arising in Deep Linear Transformers”. arXiv preprint, 2026, <https://arxiv.org/abs/2607.18584v1>

CONFERENCE PRESENTATIONS

Diverse Dynamical Behaviors in Deep Linear Transformers	SIAM MDS 2026
Diverse Dynamical Behaviors in Deep Linear Transformers	Oxford Wrkshp Math of Transformers 2026
Deriving Moments in the Age of Gossip Process from Percolation	ICASSP 2026
Information Freshness in Dynamic Gossip Networks	ITW 2025
Information Degradation and Misinformation in Gossip Network	ISIT 2025
Reconstructing Arbitrary Trees from Traces in the Tree Edit Distance Model	ISIT 2024
Age of Gossip in Random and Bi-Partite Networks	ISIT 2024
A Unified Analysis of Dynamic Interactive Learning	ALLERTON 2023

SEMINARS, TALKS, AND LECTURES

Age of information, graph recursions, and percolation.	Army Research Labs 2025
Age of information is equivalent to first passage percolation	UMD Probability Seminar 2025
Review of Universality of Perceptrons and Transformers	UMD Guest lecture Fall 2024
Percolation and Age of Gossip	Purdue CS Theory Seminar 2024
Combinatorics of the Age of Gossip Invariant	UIC CS Theory Seminar 2024
Age of Gossip in Random and Bipartite Networks	University of Maryland 2024
Kruskal, Prim, and Reverse-Delete Algorithms	Guest Lecture - UIC Algorithms 1 2023
Discrete Fourier Analysis and Intersecting Families	UIC Combinatorics Seminar 2023
Automata, Computation, and Decidability	Guest Lecture - Young Scholars Program 2022
Mathematical Modelling of Disease Spread	Guest Lecture - Young Scholars Program 2022
Combinatorial Reductions in Tree Reconstruction	UIC CS Theory Seminar 2022
String and Tree Reconstruction	UIC Graduate Student Seminar 2021

AWARDS AND FELLOWSHIPS

UIC Access To Excellence Fellow	Academic Year 2024-2025
IDEAL Institute Summer Exchange Fellow	Summer 2023

IDEAL Institute Research Fellow
INMAS Industry Fellow
TRIPODS Research Fellow
State Honors College Graduate

Spring 2023
UIUC/ R-DEX Systems Summer 2023
Spring 2021
Spring 2020

TEACHING EXPERIENCE

MCS 401 Algorithms I (Grader)	UIC Fall 2023/ Spring 2022
Young Scholars Program Instructor	UIC Summer 2022
MATH 110 Pre-Calculus (TA)	UIC Spring 2023
MCS 260 Introduction to Python (TA)	UIC Fall 2022
MCS 275 Programming Tools and File Management (TA)	UIC Fall 2021
MCS 360 Introduction to Data Structures (TA)	UIC Fall 2021/ Fall 2022
MCS 401 Algorithms I (TA)	UIC Summer 2020
MATH 165 Business Calculus (TA)	UIC Fall 2020
Quantitative Resource Center (TA)	NCF Fall 2019 – Spring 2020
Various TA Positions	NCF 2017-2019

REVIEWING

IEEE Systems Journal 2025
IEEE Journal of Selected Topics in Signal Processing 2025
ISIT 2025
SODA 2024

SERVICE

UIC MSCS Computer Science Seminar	Organizer Fall 2022
UIC MSCS Computer Science Seminar	Organizer Spring 2021
NCF ACM Chapter	Co-Founder and Organizer 2018-2020

LANGUAGES

Python, C/C++, Java, SQL, English